

10-WEEK WORK PLAN – TIC TAC TOE GAME PROJECT

Java Swing GUI Application Project Plan

Week 1: Project Planning and Research

- **Objectives:**
 - Define project objectives, scope, and target audience.
 - Research Java Swing architecture, component hierarchies, and layout management concepts.
 - Finalize the core structural blueprint of the application.

Week 2: Requirement Analysis

- **Focus:**
 - Identify layout requirements and screen navigation transitions.
 - Define operational rules for game mechanics (turn-swapping, win/draw triggers).
 - Design the overall Model-View-Controller (MVC) system architecture.

Week 3: Project Setup

- **Setup:**
 - Set up the Java development environment (Configure IDE and JDK runtime variables).
 - Create the unified core project structure and initialize the `TicTacToe.java` class scaffolding.
 - Prepare basic variable frameworks, primitive data states (`currentPlayer`, `gameStarted`), and imports.

Week 4: Home Screen Implementation

- **UI Component Building:**
 - Establish the primary application window `JFrame` dimensions and window constraints.
 - Create the landing canvas (`homePanel`) using absolute positioning layout systems (`null`).
 - Implement foundational branding typography headers (`titleLabel`) and active triggers (`startButton`).

Week 5: Game Board Configuration

- **Interface Architecture:**
 - Construct the core gameplay frame panel (`gamePanel`) using `BorderLayout` structural sections.
 - Embed a real-time status banner (`turnLabel`) to monitor current active player selections.
 - Formulate a responsive 3×3 interface grid (`boardPanel`) leveraging `GridLayout` architectures.

Week 6: Grid Integration & Rendering

- **Component Setup:**
 - Programmatically initialize the 2D matrix structure array (`JButton[3][3]`).
 - Bind customized typography fonts and structural layout configurations across all 9 board buttons.
 - Inject operational state parameters to eliminate component focusing borders (`setFocusable(false)`).

Week 7: Event Handling & Move Validation

- **Logic Development:**
 - Implement the centralized interactive event listener interface model (`ActionListener`).
 - Establish standard layout managers (`CardLayout`) to transition active cards between screens on clicks.
 - Program strict move validation rules to prevent rewriting grid data within occupied positions.

Week 8: Advanced Logic (Win & Draw Detection)

- **Algorithmic Implementation:**
 - Design the structural matrix validation engine (`checkWinner()`) targeting 8 row, column, and diagonal matches.
 - Implement grid data capacity algorithms (`isBoardFull()`) to capture terminal state draw matches.
 - Integrate instant-access control messages (`JOptionPane.showMessageDialog`) to display match end-states.

Week 9: System Integration & Optimization

- **Refinement:**
 - Combine UI panels, backend state handlers, and terminal board locking systems (`disableBoard()`).
 - Implement systematic refresh procedures (`resetBoard()`) to purge grid data states and safely restore interface variables.
 - Incorporate explicit visual markers (Distinct color codes like Blue for 'X' and Red for 'O') to improve interaction feedback.

Week 10: Finalization and Documentation

- **Project Completion:**
 - Refine component layouts, format code styles, and secure standard Object-Oriented Programming (OOP) properties.
 - Execute comprehensive system debugging cycles to address potential run-time edge cases.
 - Prepare the technical project summary documentation and compile a professional repository `README.md` file.

Summary

This 10-week work plan delivers a structured approach to building a robust Tic Tac Toe game using Java Swing. Progressing from foundational architecture and UI grid layout to logical event handling and win/draw detection, each week builds systematically upon the previous. The lifecycle ensures that the final application achieves high technical efficiency, aligns with Object-Oriented Programming (OOP) principles, and successfully completes all development deliverables within the project timeline.